



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Sequence in Metric Space

Cauchy Sequence and Completeness Theorems

MARCH 10, 2026

CONDUCTED BY

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

✉ partho.dhor@bracu.ac.bd | ✉ parthosutradhor@gmail.com

For updates subscribe on  [@ParthoSutraDhor](#)

Sequence in Metric Space

Sequence in Metric Space

Let (X, d) be a metric space. A sequence in X is a function $f : \mathbb{N} \rightarrow X$. We usually denote the sequence by $\{x_n\}$, where $x_n = f(n)$ for each $n \in \mathbb{N}$.

Convergence of a Sequence in $\mathbb{R}$

Convergence of a Sequence

A sequence $\{a_n\}$ is said to **converge** to a real number a if for every real number $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $n \geq N$, the terms of the sequence satisfy the inequality

$$|a_n - a| < \varepsilon.$$

If such a number a exists, it is called the **limit** of the sequence, and we write

$$\lim a_n = a \quad \text{or} \quad a_n \rightarrow a \text{ as } n \rightarrow \infty.$$

Convergence of a Sequence in Metric Space

Convergence of a Sequence in Metric Space

Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is said to **converge** to a point $x \in X$ if for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $n \geq N$, the terms of the sequence satisfy the inequality

$$d(x_n, x) < \varepsilon.$$

If such a point x exists, it is called the **limit** of the sequence, and we write

$$\lim x_n = x \quad \text{or} \quad x_n \rightarrow x \text{ as } n \rightarrow \infty.$$

Cauchy Sequence in Metric Space

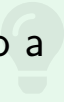
Cauchy Sequence

Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is called a **Cauchy sequence** if for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $m, n \geq N$, the terms of the sequence satisfy the inequality

$$d(x_n, x_m) < \varepsilon.$$

Every Convergent Sequence is Cauchy

Let (X, d) be a metric space. If a sequence $\{x_n\}$ in X converges to a point $x \in X$, then $\{x_n\}$ is a Cauchy sequence.



💡 Not Every Cauchy Sequence is Convergent

Let (X, d) be a metric space. It is not necessarily true that every Cauchy sequence in X converges to a point in X .

Completeness of a Metric Space

Complete Metric Space

A metric space (X, d) is called **complete** if every Cauchy sequence in X converges to a point in X .

Completeness of $\mathbb{R}^n$

💡 $\mathbb{R}$ is a Complete Metric Space

The set of real numbers $\mathbb{R}$ with the usual metric $d(x, y) = |x - y|$ is a complete metric space. 💡

💡 $\mathbb{Q}$ is not a Complete Metric Space

The set of rational numbers $\mathbb{Q}$ with the usual metric $d(x, y) = |x - y|$ is not a complete metric space. 💡

💡 $\mathbb{R}^n$ is a Complete Metric Space

The set of n -tuples of real numbers $\mathbb{R}^n$ with the usual metric



$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

is a complete metric space.

Completeness of ℓ^p Spaces

ℓ^p is a Complete Metric Space

The set of sequences ℓ^p with the metric

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{\infty} |x_i - y_i|^p \right)^{1/p} \quad \mathbf{x}, \mathbf{y} \in \ell^p$$

is a complete metric space.



💡 ℓ^∞ is a Complete Metric Space

The set of sequences ℓ^∞ with the metric

$$d(\mathbf{x}, \mathbf{y}) = \sup_{i \in \mathbb{N}} |x_i - y_i| \quad \mathbf{x}, \mathbf{y} \in \ell^\infty$$

is a complete metric space.



Completeness of Function Spaces

💡 Cauchy criterion for uniform convergence

Let $\{f_n\}$ be a sequence of functions defined on a set A and taking values in a metric space (Y, d) . The sequence $\{f_n\}$ converges uniformly to a function f on A if and only if for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $m, n \geq N$ and for all $x \in A$, the following inequality holds:

$$d(f_n(x), f_m(x)) < \varepsilon.$$

$C[0, 1]$ under sup metric is a Complete Metric Space

Let $C[0, 1]$ be the set of all continuous real-valued functions defined on the interval $[0, 1]$. Define a metric d on $C[0, 1]$ by

$$d(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)|$$

for all $f, g \in C[0, 1]$. Show that $(C[0, 1], d)$ is a complete metric space.

❓ $C[0, 1]$ under integral metric is not Complete

Let $C[0, 1]$ be the set of all continuous real-valued functions defined on the interval $[0, 1]$. Define a metric d on $C[0, 1]$ by

$$d(f, g) = \int_0^1 |f(x) - g(x)| dx$$

for all $f, g \in C[0, 1]$. Show that $(C[0, 1], d)$ is not a complete metric space.

Completeness and Closed Sets

💡 Closure of a Set and Convergence of Sequences

Let (X, d) be a metric space and let A be a subset of X . A point $x \in X$ is in the closure of A if and only if there exists a sequence $\{a_n\}$ in A that converges to x .

Closed subspace of a Complete Space is Complete

Let (X, d) be a complete metric space and let (Y, d) be a subspace of X . Then the metric space (Y, d) is complete if and only if Y is closed in X .

Cantor's Intersection Theorem

💡 Cantor's Intersection Theorem

Let (X, d) be a complete metric space. If $\{F_n\}$ is a sequence of non-empty closed subsets of X such that $F_{n+1} \subseteq F_n$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} \text{diam}(F_n) = 0$, then the intersection $\bigcap_{n=1}^{\infty} F_n$ contains exactly one point.

Thank You!

We'd love your questions and feedback.

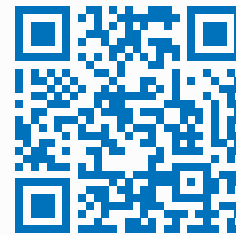
Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

✉ partho.dhor@bracu.ac.bd | ✉ parthosutradhor@gmail.com

 **@ParthoSutraDhor**

(Lectures, walkthroughs, and course updates)



Scan for the channel

References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.